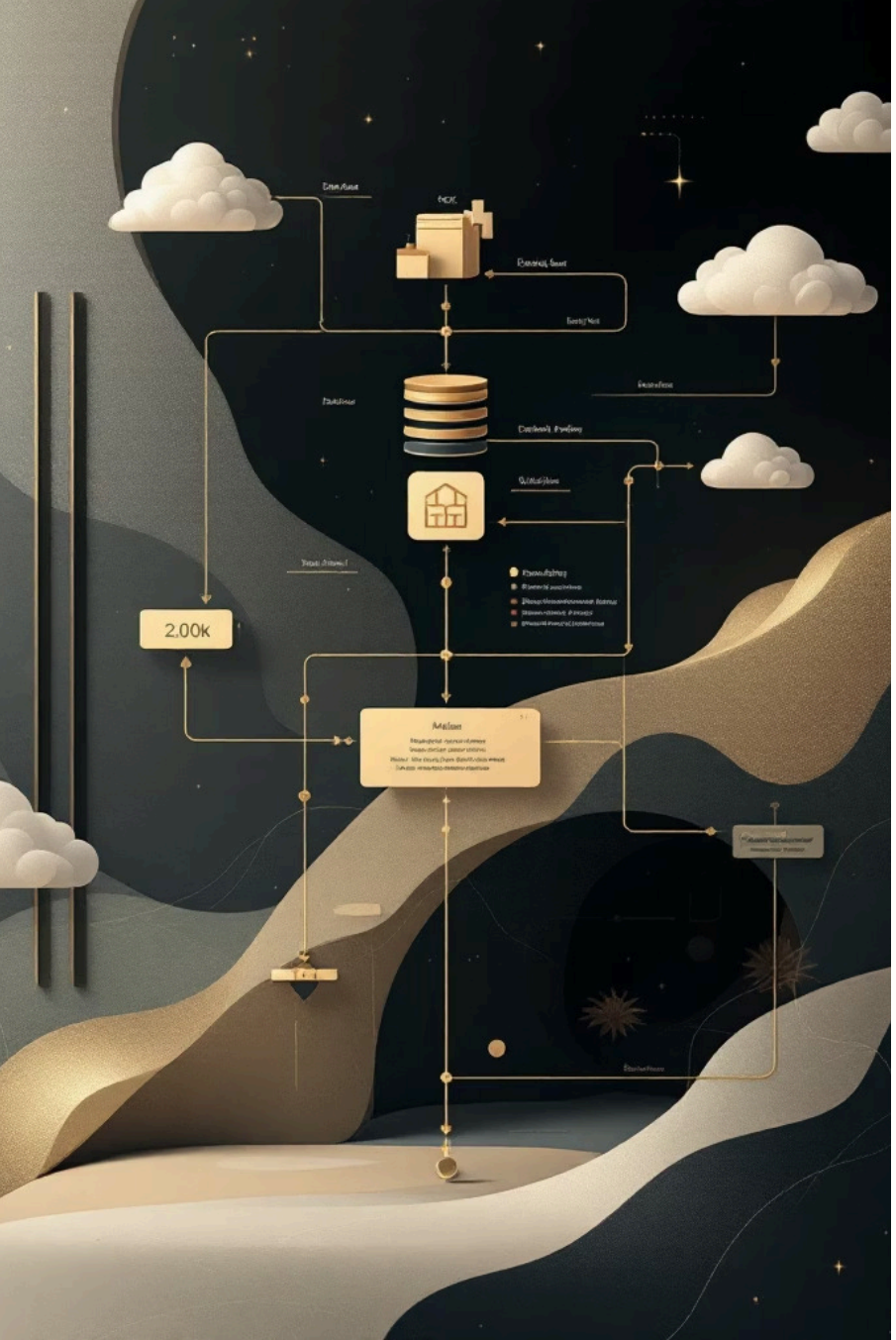
Iris Flower Classification

Decision Tree Machine Learning Project

Made with GAMMA



Project Overview

Goal

Classify iris species using physical features

Dataset

150 samples with 4 features from Scikit-learn

Species

Setosa, Versicolor, Virginica

Made with GAMMA

Dataset Features

Physical Measurements

- Sepal Length
- Sepal Width
- Petal Length
- Petal Width

Made with GAMMA

Tools & Technologies

Python

Core programming language

Pandas & NumPy

Data manipulation

Matplotlib & Seaborn

Data visualization

Scikit-learn

ML algorithms

Streamlit

Web interface

SQL & Power BI

Data handling

Made with GAMMA

Project Workflow

01

Load Dataset

Import from Scikit-learn

02

Data Exploration

Analyze structure and distribution

03

Visualization

Pairplot relationships

04

Train-Test Split

80% training, 20% testing

05

Model Training

Gini & Entropy criteria

06

Evaluation

Accuracy, classification report

07

Confusion Matrix

Heatmap visualization

08

Tree Visualization

Decision rules display

09

Prediction

Test new data

Made with GAMMA

Model Training

Decision Tree (Gini)

- Uses Gini impurity
- Faster computation
- Efficient splitting

Decision Tree (Entropy)

- Uses Information Gain
- More informative splits
- Better interpretability

Made with GAMMA

Model Evaluation

Accuracy Score

Overall correctness of predictions

Classification Report

Precision, recall, F1-score

Confusion Matrix

Predicted vs actual values

Made with GAMMA

Results

100%

Accuracy

Perfect classification achieved

3

Species

Setosa, Versicolor, Virginica

150

Samples

Complete dataset analyzed

Both models performed with high accuracy. Entropy-based tree provided slightly better interpretability with clear class separation observed.

Made with GAMMA

Key Learnings

Decision Tree Algorithms

Understanding Gini vs Entropy criteria

Model Evaluation

Techniques for assessing performance

Data Visualization

Seaborn for insightful plots

Feature Importance

Understanding physical characteristics

Real-World ML Workflow

Complete project implementation

Made with GAMMA

Future Improvements

Hyperparameter Tuning

Max depth, pruning techniques

Cross-Validation

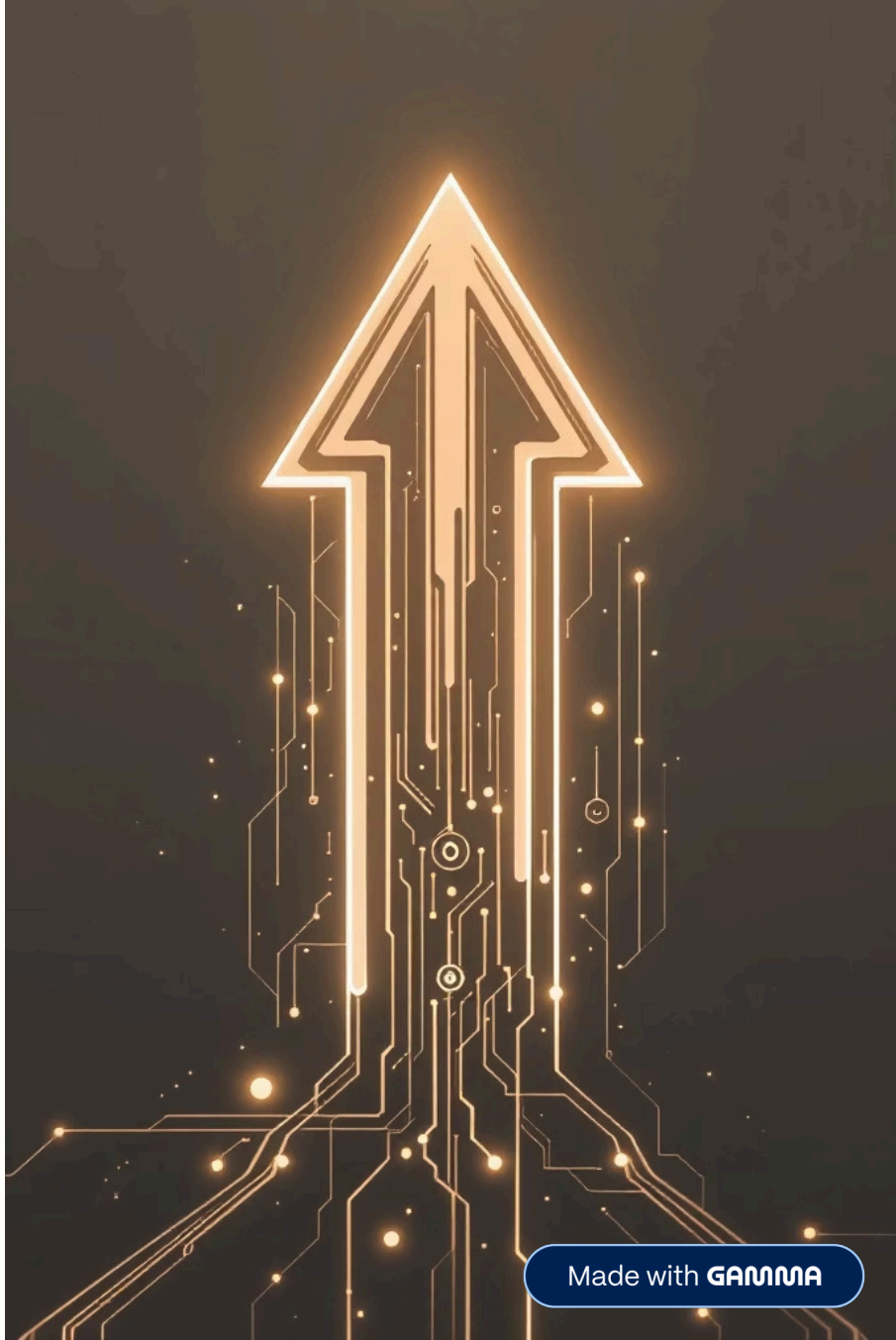
Robust model evaluation

Streamlit Deployment

Interactive web application

Model Comparison

SVM, Random Forest



Made with GAMMA